

ACUTE APNEA SWIMMING: METABOLIC RESPONSES AND PERFORMANCE

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ABSTRACT

Guimard, A, Prieur, F, Zorgati, H, Morin, D, Lasne, F, and Collomp, K. Acute apnea swimming: Metabolic responses and performance. *J Strength Cond Res* 28(4): 958–963, 2014—Competitive swimmers regularly perform apnea series with or without fins as part of their training, but the ergogenic and metabolic repercussions of acute and chronic apnea have not been examined. Therefore, we aimed to investigate the cardiovascular, lactate, arterial oxygen saturation and hormonal responses to acute apnea in relation to performance in male swimmers. According to a randomized protocol, 15 national or regional competitive swimmers were monitored while performing four 100-m freestyle trials, each consisting of four 25-m segments with departure every 30 seconds at maximal speed in the following conditions: with normal frequency breathing with fins (F) and without fins (S) and with complete apnea for the four 25-m segments with (Fap) and without fins (Sap). Heart rate (HR) was measured continuously and arterial oxygen saturation, blood, and saliva samples were assessed after 30 seconds, 3 minutes, and 10 minutes of recovery, respectively. Swimming performance was better with fins than without both with normal frequency breathing and apnea ($p < 0.001$). Apnea induced no change in lactatemia, but a decrease in arterial oxygen saturation in both Sap and Fap ($p < 0.001$) was noted and a decrease in HR and swimming performance in Sap ($p < 0.01$). During apnea without fins, performance alteration was correlated with bradycardia ($r = 0.63$) and arterial oxygen desaturation ($r = -0.57$). Saliva dehydroepiandrosterone was increased compared with basal values whatever the trial ($p \leq 0.05$), whereas no change was found in saliva cortisol or testosterone. Further studies are necessary to clarify the fin effect on HR and performance during apnea swimming.

KEY WORDS swimmers, heart rate, pulse oximetry, lactates, steroid hormones

INTRODUCTION

Apnea induces cardiovascular adaptations, called diving responses (1–4). The cardiac output decreases because of concurrent reductions in stroke volume and heart rate (HR) (5,9,10,15,23,25). Peripheral blood flow reduces because of arterial vasoconstriction, whereas blood flow in the carotid artery increases (11,19), suggesting redistributed blood flow toward vital organs during apnea. The purpose seems to be an oxygen-conserving effect with an overall oxygen consumption decrease (7). Arterial oxygen desaturation occurs in parallel and plasma lactate concentration is increased from baseline after apnea, suggesting greater reliance on anaerobic metabolism during breath-holding (1,10,12,18). As a result of the stress caused by breath-hold diving, catecholamine blood concentration was found to be increased compared with control conditions (16). Unfortunately, the responses of other stress hormones: cortisol, dehydroepiandrosterone (DHEA), and testosterone have never been investigated. Diving responses seem to be augmented by several factors, such as face immersion in cold water and hypoxia, resulting in a deeper nadir in arterial oxygen saturation, and greater lactatemia (2).

Studies of the metabolic responses to apnea during dynamic exercise remain relatively scarce. It was found that the diving-induced bradycardia during moderate dynamic cycle exercise was powerful (7) enough to override the exercise tachycardia for the period of apnea (1–4,24). However, more recent studies in elite synchronized swimmers (22) and trained breath-hold divers (17) showed that with higher exercise intensity, HR was increased during dynamic apnea compared with dry static apnea, but a relative bradycardic response was still observed during the dynamic apnea. Arterial oxygen desaturation occurred more quickly in trained divers during maximal dynamic breath-hold dives compared with static dives and, as expected, the postdive lactate increase was greater for the combination of apnea and dynamic exercise (7). Other studies have shown a relationship between the bradycardic response or arterial oxygen desaturation and performance in elite synchronized swimmers, divers, and underwater hockey players (7,14,18,22).

In modern swimming, athletes regularly perform swimming series in apnea with or without fins as part of their training. This type of training can increase the underwater glide distance and

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it probably improves aerobic and/or anaerobic capacity, as well (13). However, no field or laboratory study has yet confirmed these potential benefits of apnea training. Moreover, to our knowledge, the metabolic and ergogenic repercussions of acute apnea have never been examined in swimming.

This study assessed the cardiovascular, lactate, arterial oxygen saturation, and hormonal responses (i.e., cortisol, DHEA, and testosterone) to acute apnea in relation to performance in young male competitive swimmers of regional or national level.

METHODS

Experimental Approach to the Problem

This investigation used a 4 modality repeated measure design to compare performance (swim time) and metabolic responses (HR, blood lactate, arterial desaturation, steroid hormones) in

trained swimmers with and without apnea, and with and without fins. Most previous studies were conducted during dives and demonstrated that cardiac output was reduced throughout apnea during exercise, essentially because of bradycardia, whereas systemic resistance was increased. In parallel, both static and dynamic apneas induced marked arterial desaturation and lactatemia increases. However, to our knowledge, this is the first study that measured the physiological responses to apnea in swimmers. It was hypothesized that the simultaneous combination of acute apnea and high-intensity exercise would alter performance and metabolic responses in our swimmers with and without fins.

Subjects

The study was carried out in accordance with the declaration of Helsinki and was approved by the Institutional

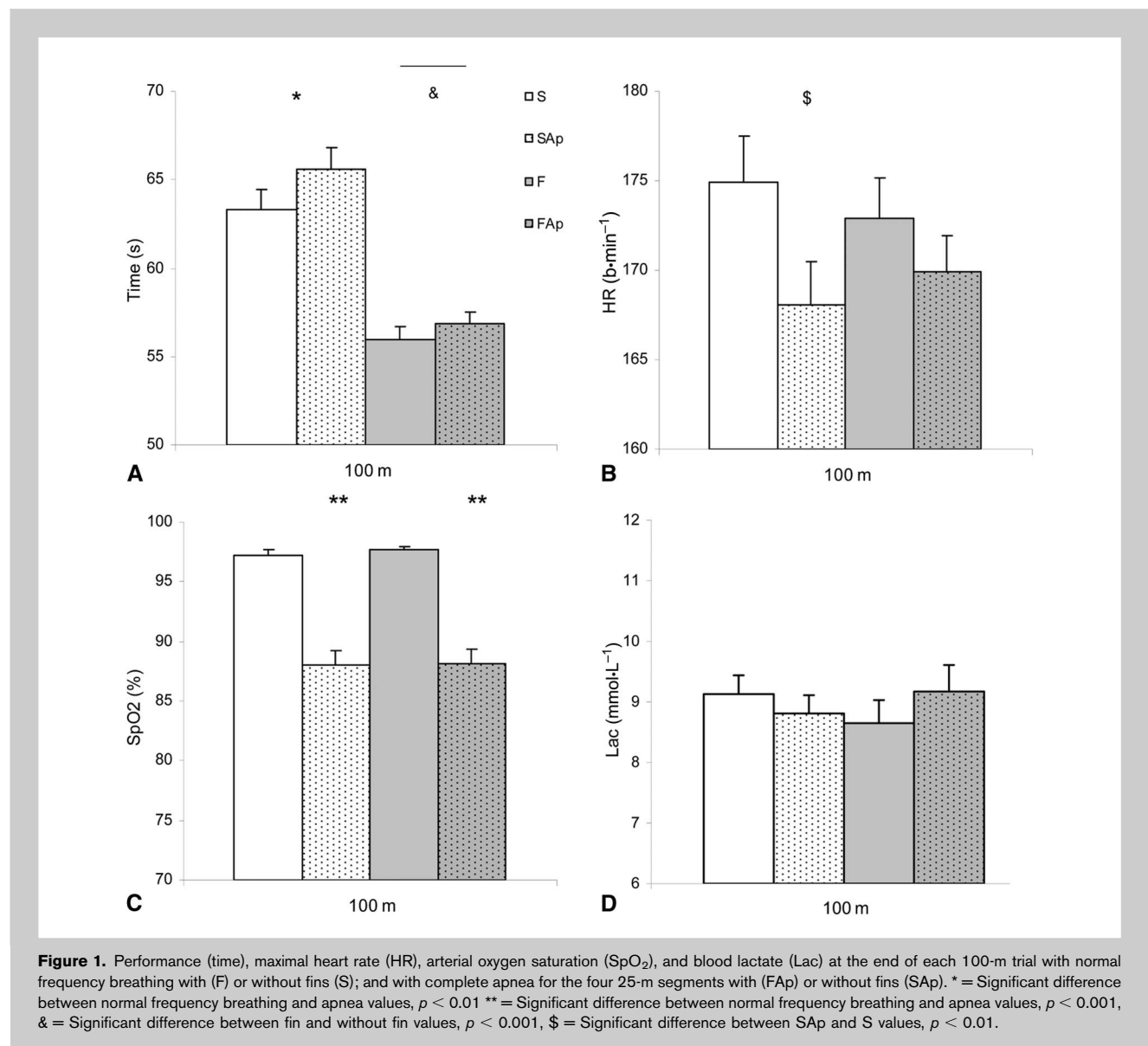


TABLE 1. Saliva cortisol, DHEA, and testosterone concentrations at the end of each 100-m trial with normal frequency breathing with (F) and without fins (S), and with complete apnea during all four 25-m segments with (FAp) and without fins (SAp).

Steroid hormones	Rest	S	SAp	F	FAp
Cortisol (ng·ml ⁻¹)	1.3 ± 0.2	1.5 ± 0.2	2.1 ± 0.4	2.0 ± 0.3	2.2 ± 0.4
DHEA (pg·ml ⁻¹)	221.3 ± 17.8	347.3 ± 31.2*	349.1 ± 40.1*	338.6 ± 30.4*	381.2 ± 31.7*
Testosterone (ng·ml ⁻¹)	56.8 ± 2.9	54.2 ± 3.2	50.7 ± 3.2	52.2 ± 3.0	63.1 ± 5.0

*Significant difference compared with rest values, $p \leq 0.05$.

Review Board of the University. Fifteen young trained regional- to national-level competitive male swimmers, 5–12 years of swimming, 3–7 times per week (Age, 21.9 ± 0.9 years; Height, 180.1 ± 2.0 cm; Weight, 69.5 ± 1.7 kg) volunteered to participate in the study and gave written consent after being informed of the nature of the experiments. None of them was taking any medication or had a family history of any endocrine disorder.

Procedure

Trials were held at the same time of day (10 AM–12 PM) for each subject to prevent diurnal variations in hormonal responses. On the actual testing day, subjects reported to the swimming pool between 9 AM and 10 AM, 2 hours after ingesting a small meal (about 500 kcal), which was confirmed through self-reported diet records and questioning. The swimmers began with a 15-minute standardized warm-up followed by 15 minutes of passive recovery and then swam a total of four 100-m freestyle trials at maximal speed in randomized order, separated by 30-minute recoveries. Each 100-m freestyle trial consisted of four 25-m segments with departure every 30 seconds. The conditions were as follows:

(a) at normal frequency breathing without fins (S), (b) with complete apnea for the four 25-m segments without fins (SAp), (c) at normal frequency breathing with standard commercial fins (F), and (d) with complete apnea for the four 25-m segments with fins (FAp). Nabaiji fins (Decathlon) were used in this study. These fins were made of rubber, small in size and highly flexible. Water was given ad libitum during exercise.

Swimming performance was assessed by the 100-m freestyle time. For the metabolic parameters, HR was continuously

measured during the 4 trials with waterproof HR monitors (RCX5 Polar; Polar Electro Oy, Kempele, Finland). Arterial saturation was measured with a Nonin Onyx 9590 Finger Pulse Oximeter (Nonin Medical, Minnesota, USA) at rest and 30 seconds after the end of the last 25 m of each trial. For lactatemia measurement (Lactate pro, Arkray, Kyoto, Japan) and cortisol, DHEA, and testosterone analyses, 10 μ l of capillary blood from the fingertip and 1.5 ml of saliva were also drawn at rest and, respectively, 3 and 10 minutes after the end of the last 25 m of each trial. The unstimulated saliva was collected by the subjects themselves using Salitubes (DRG Diagnostic, Marburg, Germany). No sample was collected from an athlete with oral disease, inflammation, or lesion. The Salitubes were promptly stored at 4° C within the hour and at –20° C within the 3 days until analysis. Each sample had to be frozen, thawed, and centrifuged at least once to separate the mucins. Enzyme-linked immunosorbent assays (ELISA) were used for the saliva cortisol, DHEA, and testosterone analyses (DRG Diagnostics). Assays were made in duplicate and coefficients of variation for all parameters were always <10%.

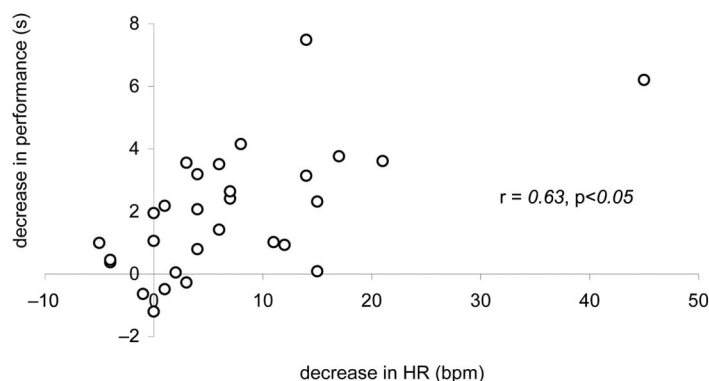


Figure 2. Correlation during apnea swimming between the decrease in performance (expressed in seconds) and the decrease in heart rate (expressed in b·min⁻¹).

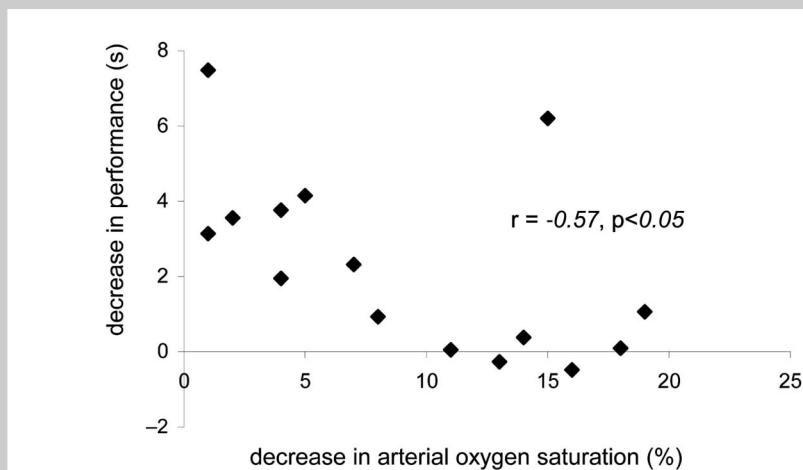


Figure 3. Correlation during apnea swimming between the decrease in performance without fins (expressed in seconds) and the arterial oxygen saturation (expressed in percentage).

Statistical Analyses

Data are presented as mean values $\pm$ SE. After verifying the assumption of normality, analyses of variance (ANOVA) with repeated measures were used to compare differences in all dependent variables between trials, and a post-hoc Tukey test was used to locate the differences, if necessary. The limit of significance was $p \leq 0.05$. Correlations were calculated using Pearson's product-moment correlation test. The null hypothesis was rejected at $p \leq 0.05$.

RESULTS

Performance (Time)

Swimming time was lower with fins than without fins, with or without apnea ($p < 0.001$), with an increase in velocity (V) of about $0.2 \text{ m} \cdot \text{s}^{-1}$: V_S , 1.59 ± 0.11 ; V_{SAp} , $1.54 \pm 0.12 \text{ m} \cdot \text{s}^{-1}$ compared with V_F , 1.80 ± 0.10 ; V_{FAP} , $1.77 \pm 0.08 \text{ m} \cdot \text{s}^{-1}$ ($p < 0.001$). Apnea swimming induced an alteration in performance without fins ($p < 0.01$), but not with fins (Figure 1).

Basal SpO_2 did not differ between trials (98.2–98.5%). Apnea swimming induced a decrease in SpO_2 both with and without fins ($p < 0.001$, Figure 1).

Heart Rate

Maximal HR decreased in SAp compared with S and F (respectively, $p < 0.01$ and $p \leq 0.05$). However, there was no difference in maximal HR between the SAp and FAp conditions.

Blood Lactate

End exercise blood lactate was not different between trials (Figure 1).

Hormonal Analyses

Cortisol and testosterone values were quite similar across trials and showed no increase compared with basal values.

However, end exercise DHEA values were higher than rest values for all trials ($p \leq 0.05$, Table 1).

Correlations

There was a positive correlation between the alteration in performance (difference between normal breathing and apnea performance times expressed in seconds) and the exercise bradycardia induced by apnea (difference between normal breathing and apnea HR expressed in $\text{b} \cdot \text{min}^{-1}$), both without fins and with fins ($r = 0.63$, $p \leq 0.05$, Figure 2). In parallel, a negative correlation was observed only without fins between the alteration in performance

and arterial oxygen saturation ($r = -0.57$, $p \leq 0.05$, Figure 3).

DISCUSSION

Apnea training was empirically identified as one of the factors that might contribute to improving the performances of elite swimmers. Indeed, apnea durations clearly increase with training (14) and in addition, some authors hypothesize that apnea training may be an effective alternative to hypobaric or normobaric hypoxia to increase aerobic or anaerobic performance (13). However, there is very little scientific evidence of its impact. Moreover, to our knowledge, no previous study has examined the physiological and ergogenic responses to either acute or chronic apnea in swimmers. The present study shows that acute dynamic apnea swimming elicits a decrease in arterial oxygen saturation, HR, and swimming performance in regional- and national-level male swimmers. However, the decrease in HR and swimming performance induced by apnea disappears with fin use.

As expected, performance was better at normal frequency breathing, with faster swimming times (around 7 seconds for 100 m) with fins vs. without fins, as fins provide thrust to overcome drag and propel the swimmer (20,21). At comparable speeds, Zamparo et al. (26) reported that the energy cost when swimming with fins was about 40% lower than when swimming without them; however, when compared at the same metabolic power, the decrease in energy cost allowed an increase in velocity of about $0.2 \text{ m} \cdot \text{s}^{-1}$. Our increase in velocity with fins was similar, and, in view of the lack of change in HR, lactatemia, arterial oxygen saturation, or stress hormone concentrations between the 2 conditions, it may be suggested that the trials with and without

fins have been performed at the same metabolic power. Direct measure of oxygen consumption is of course needed to confirm this hypothesis.

Heart rate response to acute swimming apnea confirmed the findings of earlier studies of bradycardic response to apnea (1–5,9,10,15,23,25). Bradycardia in water does not depend on the diving depth, but rather on the length of the breath-hold (6), with change further accentuated by cooling of the facial area (1,11,23). During high-intensity exercise, like that performed in the present study, cardiovascular demands are greater and HR is much faster than with dry static apnea. However, we still observed a relative bradycardic response during acute swimming apnea compared with normal frequency breathing conditions in our swimmers at the end of the 100 m (around $10 \text{ b} \cdot \text{min}^{-1}$). These data are, in accordance with, earlier reports in elite synchronized swimmers, trained divers, and underwater hockey players performing dynamic apnea (14,16,17,22). In parallel, we observed a decrease in arterial oxygen saturation (around 10%) during the swimming apnea trials. This finding is also in accordance with the literature and probably reflects a reduction in skeletal muscle saturation that starts earlier than the reduction in arterial saturation (18). However, contrary to previous studies conducted in trained apneic divers (12,23), lactatemia was not higher with acute apnea compared with control conditions in our swimmers. One explanation may be that the duration of 25-m bouts of swimming apnea was too short to increase the reliance on anaerobic glycolytic metabolism. Another hypothesis is that the lactatemia peak may have occurred later than in control conditions (7) because of compromised removal from working muscle in apnea because of vasoconstriction (7). Moreover, restricted blood flow may lead to considerable regional differences in lactate concentration (22). Further studies seem necessary to clarify this point and to understand the mechanisms that are implicated.

The stress induced by apnea swimming did not result in differences in the hormone concentrations. The exercise, DHEA increase was identical in all conditions, with no change in cortisol and testosterone compared with basal conditions. Nevertheless it is possible that this lack of change may have been because of the short duration of the trial and that longer or chronic apnea swimming would increase these stress hormones, as seen with erythropoietin (8).

It is interesting to note that the alteration in swimming performance with apnea was positively correlated with exercise bradycardia and negatively with the arterial oxygen desaturation without fins. It therefore seems that performance is directly linked to the subject's capacity to supply oxygen to exercising muscles. This result is partially in accordance with the literature. Rodriguez-Zamora et al. (22) showed that the magnitude of the bradycardic response in elite synchronized swimmers seemed to be related to performance variability, whereas Palada et al. (18) demonstrated that a greater reduction in oxygenated hemoglobin occurred in breath-hold divers compared with nondivers, because of

longer breath-hold duration. However, direct comparison with these earlier studies is difficult, because (1) our swimmers were not apnea-trained and (2) the physiological and technical demands of the exercises in these studies showed wide disparities.

The results obtained with fins were unexpected. One might assume comparable repercussions of acute apnea swimming on performance and metabolism with and without fins. . However, although we observed the same arterial oxygen desaturation with and without fins in apnea swimming, the decrease in HR and swimming performance observed in apnea without fins disappeared with fins. It seems unlikely, but not impossible that the lack of decrease in these parameters was linked to the shorter apnea time with fins. The most probable hypothesis was, however, that the fins had a stimulating influence in HR, because of the more powerful work of the legs with finswimming, directly helping the maintenance of the cardiac output and performance. Further investigations are specifically needed (a) to further examine the specific responses related to finswimming, (b) to determine the effects of acute apnea swimming of different durations, particularly on lactate and hormonal responses, and (c) to measure the physiological adaptations to standardized training in apnea swimming and the impact on performance.

PRACTICAL APPLICATIONS

An increasing number of coaches consider apnea training as a new training method that might contribute to improving the performances in sports like diving, underwater hockey and rugby, synchronized swimming, finswimming, and swimming. Indeed, apnea training increases apnea duration and in addition, it has been suggested that apnea training may be an effective alternative to hypobaric or normobaric hypoxia to increase aerobic or anaerobic performance. However, no field or laboratory study has yet confirmed these potential benefits of apnea training in swimming or finswimming, and the performance and metabolic repercussions of acute apnea have never been examined. The key finding of the present work is that short bouts of acute apnea swimming at high intensity alter maximal HR, arterial oxygen saturation, and the performance. These data seem consistent with previous findings conducted in breath-hold divers and synchronized swimmers, although it is difficult to compare results because of the wide discrepancies in terms of apnea duration, exercise intensity, and technique in the range of sports performed. Moreover, it is important to note that the decrease in HR and swimming performance observed in the present study during apnea without fins disappears with fin use, suggesting that fins may directly help to maintain the cardiac output needed for oxygen supply.

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